USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LESSON 21 LAB - Antenna Pattern
Lab Sheet
Fall 2026

Name: _____ **Section:** _____ **Date:** _____

This sheet is the turn-in document for the L21 Lab. Report every lobe amplitude relative to the main-lobe peak, in dBc, and state the uncertainty next to any number read off a hand-walked trace. The lab is individual: one sheet per cadet, from your own kit.

Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

1. **The annotated trace.** Attach or sketch one screenshot of your best broadside run from the Tracking tab, with the main lobe and both first sidelobes labeled with their dBc values.

2. **The amplitude comparison table.** Three lobes from the broadside run, each as an amplitude relative to the main-lobe peak. The mechanical column is your hand-walked trace, the electrical column is the Rectangular-tab sweep of step B4, and the calculated column comes from theory.

Lobe	**Mechanical (dBc)**	**Electrical sweep (dBc)**	**Calculated (dBc)**	**Mechanical − electrical**
Main lobe				
First sidelobe, left				
First sidelobe, right				

3. **Written answers.** Three or four sentences each.

   (a) Why do the lobe amplitudes survive a sloppy rotation while the lobe angles do not? Refer to what the time axis does and does not record.

   (b) Which of the four artifacts of the hand measurement does an anechoic chamber remove, and which of them would still be present in a chamber measurement?

Documentation: _____